

WHAT IS PROOF-OF-CONCEPT?

What is proof-of-concept (PoC)? Is it demonstration of working videos of the projects published? How can I watch these videos?

Prabhat Kumar

EFY. EFY has been presenting the proof-of-concept (PoC) videos of the DIY projects published in Electronics For You magazine since September 2021 to demonstrate the final output of those projects. To view these videos, please check <https://www.electronicsforu.com/category/electronics-projects/diyprojects-video-tutorials>

• • •

WANT TO PUBLISH PROJECTS IN EFY

I am hobbyist and circuit designer and I want to publish my DIY projects in Electronics For You magazine. Do you have any guidelines for the contributors of such articles?

Raghav

EFY. Circuit designers are welcome to contribute DIY projects as well as other technical articles. EFY pays an honorarium for the original articles that are published on EFY website(s) or Electronics For You magazine. The guidelines for contributors can be found at <https://www.electronicsforu.com/news/what-is-new-at-electronicsforu-updates/faq-authors-contributors-efy>

• • •

MIND-CONTROLLED PROSTHETIC ARM

This is regarding the project 'Mind-Controlled Prosthetic Arm' published in January issue. Please guide me with the source for procuring the EEG sensor and the source code for the project.

Yogesh Kumar

EFY. You can buy the sensor from online electronics stores. The code for

COMPARISON OF TRANSISTORS			
Type	Bipolar transistors	MOSFETs	IGBTs
Gate (base) drive	Current drive (Low input impedance)	Voltage drive (High input impedance)	Voltage drive (High input impedance)
Gate (base) drive circuit	Complicated for switching applications	Relatively simple	Relatively simple
On-state voltage characteristics	Low VCE(sat)	On-resistance x drain current Without built in voltage	Low VCE(sat) With built-in voltage
Switching time	Slow (Carrier accumulation effect)	Ultra-high speed (Unipolar device)	High speed (Faster than bipolar transistors and slower than MOSFETs)
Parasitic diode	Not present	Present (body diode)	Present only in RC-IGBTs

the project is available at <https://www.electronicsforu.com/electronics-projects/mind-controlled-prosthetic-arm>

• • •

SMART WIRELESS LOCKING SYSTEM

I need software program for the project 'OTP Based Smart Wireless Locking System Using Arduino' that was published recently. Where can I find it?

Krishnan

EFY. You can download the code using the link <https://www.electronicsforu.com/wp-admin/post.php?post=51716&action=edit>

• • •

IGBT VS MOSFET

Is it possible to use an IGBT in place of a power MOSFET in a project?

Sanjiv Mohbansi

EFY. There are three major types of transistors available: bipolar transistors, MOSFETs, and IGBTs. The table above compares the performance and characteristics of these transistors.

Bipolar transistors are now hardly ever used for power electronics and switching applications because of the need for drive and protection circuits and slow switching speed. Instead, MOSFETs and IGBTs are selectively used according to the required characteristics.

In the low-current region, the MOSFET exhibits a lower on-state voltage than the IGBT. However, in

the high-current region, the IGBT exhibits lower on-state voltage than the MOSFET, particularly at high temperature. IGBTs are commonly used at a switching frequency lower than 20kHz, because they exhibit higher switching loss than unipolar MOSFETs.

• • •

PROGRAMMING CODE

Please assist me with the programming code of AVR machine using Atmega 16A.

Anthony

EFY. There are several ways to program AVR microcontrollers. You can program Atmega16 using ISP programmer. The software is written in C language and compiled using Keil software. You can use any suitable software for programming the ATmega16A microcontroller. For instance, ProgISP programmer can be used for its programming.

You can install ProgISP from the internet. After successful installation, follow the steps to burn (program) the code into Atmega ICs using ProgISP programmer:

1. Install ProgISP
2. Save source code in hex code
3. Open ISP programmer
4. Select IC here as Atmega16A and insert the IC in zip socket
5. Upload the hex code
6. The window shows: Uploaded Successfully
7. Remove the IC and use it for the project

Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: onedit@efy.in)